and pressure with V_i and V_f as the total volumes of the reactant and product gases respectively, and n_i and n_f as the number of moles of gaseous reactants and products, then,

For reactants (initial state) :

$$PV_i = n_i RT \text{ --------- (7.14)}$$

For products (final state) :

$$PV_f = n_f RT \text{ --------- (7.15)}$$

(7.15) - (7.14)

$$P(V_f - V_i) = (n_f - n_i) RT$$

$$P\Delta V = \Delta n_{(g)} RT \text{ --------(7.16)}$$

Substituting in 7.16 in 7.12

$$\Delta H = \Delta U + \Delta n_{(g)} RT \text{--------(7.17)}$$

7.5.2 Enthalpy Changes for Different Types of Reactions and Phase Transitions:

The heat or enthalpy changes accompanying chemical reactions is expressed in different ways depending on the nature of the reaction. These are discussed below.

Standard heat of formation

The standard heat of formation of a compound is defined as "the change in enthalpy that takes place when one mole of a compound is formed from its elements, present in their standard states (298 K and 1 bar pressure)". By convention the standard heat of formation of all elements is assigned a value of zero.

$Fe(s) + S(s) \rightarrow FeS(s)$

$$\Delta H_f^0 = -100.42 \text{ kJ mol}^{-1}$$

$2C(s) + H_2(g) \rightarrow C_2H_2(g)$

$$\Delta H_f^0 = +222.33 \text{ kJ mol}^{-1}$$

$$\frac{1}{2} Cl_2(g) + \frac{1}{2} H_2(g) \rightarrow HCl(g)$$

$$\Delta H_f^0 = -92.4 \text{ kJ mol}^{-1}$$

The standard heats of formation of some compounds are given in Table 7.4.

Table: 7.4 standard heat of formation of some compounds

Substance	ΔH_f^0 (kJ mol^{-1})	Substance	ΔH_f^0 (kJ mol^{-1})
$H_2O(l)$	−242	$CH_4(g)$	-74.85
HCl(g)	−92.4	$C_2H_6(g)$	−84.6
HBr(g)	−36.4	$C_6H_6(g)$	+49.6
NH3(g)	−46.1	$C_2H_2(g)$	+222.33
$CO_2(g)$	-393.5	$CH_3OH(l)$	−239.2

7.6 Thermochemical Equations:

A thermochemical equation is a balanced stoichiometric chemical equation that includes the enthalpy change (ΔH). The following conventions are adopted in thermochemical equations:

(i) The coefficients in a balanced thermochemical equation refer to number of moles of reactants and products involved in the reaction.

11th Std Chemistry 186-228.indd 198 23-12-2021 15:33:47